

REPORT CONTENT SHEET

Raw material for your futuristic report —
Build It in English

HABIT LOOP

H A B I T T R A C K E R



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Project /
Proyecto

HABIT LOOP — Habit Tracker



Pitch

A lightweight PHP web application to build,
track, and maintain daily habits effortlessly.



Environment /
Entorno

PHP 8.x + Apache (XAMPP)



Report medium /
Medio del informe

Word exported to PDF



Date / Fecha

2026-09-01



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2. The environment used

DEVELOPMENT ENVIRONMENT



PHP is lightweight, easy to deploy locally with XAMPP, and native to server-side script execution. It allows simple file handling (JSON) or database queries (MySQL) without complex configuration overhead, making it perfect for rapid habit tracking logic.

PHP es ligero, fácil de ejecutar localmente con XAMPP y nativo para scripts de servidor. Permite gestionar archivos JSON o bases de datos MySQL de forma sencilla y sin configuraciones complejas, lo que lo hace ideal para desarrollar la lógica del rastreador de hábitos rápidamente.

2.1 Alternative not used

CONSIDERED TECHNOLOGY



Node.js was considered, but skipped because it requires managing NPM packages and asynchronous runtime setups that add unnecessary complexity for a simple habit tracking project.

Se consideró Node.js, pero se descartó porque requiere administrar paquetes de NPM y configuraciones asíncronas que añaden complejidad innecesaria para un proyecto simple de seguimiento de hábitos.

EXPLORE
DEVELOP
CREATESIMPLE
TOOLS
BIG IDEAS

3. Tools

DEVELOPMENT TOOLS



TOOL	VERSION	WHAT IT WAS USED FOR (ENGLISH)
Visual Studio Code	1.85.0	Writing and editing backend PHP code, HTML structure, and CSS styling.
XAMPP	8.2.12	Running the local Apache web server and PHP execution environment.
Visual Studio Code	Latest	Testing application responsiveness, UI layout, and browser debugging.
Git	2.43.0	Managing source code version control and tracking project changes.



THE RIGHT TOOLS
TURN IDEAS INTO
REAL SOLUTIONS

PLAN
DEVELOP
LEARN
IMPROVE*Better
Tools
Brighter
Ideas*

4. Resources

TOOLS THAT MADE IT POSSIBLE

RESOURCE TYPE	DETAIL	CONTRIBUTION TO THE PROJECT
 Hardware	Laptop Intel Core i5, 8GB RAM	Provided the primary development environment for writing code and testing locally.
 Time invested	12 hours	Divided between Sprint 0 charter preparation, backend logic, and UI design.
 Documentation / tutorials	PHP Official Documentation & W3Schools	Consulted for syntax validation, date handling methods, and file system logic.
 Libraries or packages	Bootstrap 5 / FontAwesome	Provided basic layout styling components and UI icons for habit status.
 AI or assistance used	ChatGPT / Gemini	Assisted in structuring technical documentation, drafting English microcopy, and code debugging.

THE RIGHT RESOURCES
TURN IDEAS INTO **REAL SOLUTIONS**

5. Functionality

HOW IT WORKS FOR THE USER

USER EXPERIENCE IN ACTION

*Simple
Actions
Real
Results*

FEATURE	WHAT THE USER DOES AND SEES (ENGLISH)
 Habit Creation	The user clicks "+ Add New Habit", fills in the name and weekly target, and sees the new habit listed on the main dashboard.
 Daily Check-in	The user clicks the checkbox next to a habit. The status updates instantly to completed and increments the streak counter.
 Progress Tracking	The user opens the progress view and sees a weekly percentage chart reflecting their completed target days.
 Habit Deletion	The user clicks the delete button on a habit, confirms the dialog, and sees the habit removed from the list.



5.1 Out of scope

CURRENT LIMITATIONS

The application does not support multi-user authentication, cloud synchronization across devices, push notifications outside the browser session, or social sharing features.

La aplicación no incluye autenticación de múltiples usuarios, sincronización en la nube entre diferentes dispositivos, notificaciones push fuera del navegador, ni funciones para compartir en redes sociales.

6. Architecture and data model

SYSTEM STRUCTURE



The application follows a simple Modular MVC structure. The `/public` directory holds entry points (`index.php`) and static assets (`/css`, `/js`). Core application logic resides in `/src`, featuring `Habit.php` for data models and `HabitController.php` for handling actions like adding or completing habits. The `/config` folder contains database setup scripts and global helper functions.

La aplicación sigue una estructura Modular MVC sencilla. El directorio `/public` contiene el punto de entrada (`index.php`) y archivos estáticos (`/css`, `/js`). La lógica principal está en `/src`, con `Habit.php` para los modelos de datos y `HabitController.php` para gestionar acciones como agregar o completar hábitos. La carpeta `/config` incluye la configuración de la base de datos y funciones globales.

Project Structure



`/public`

- `index.php`
- `/css`
- `/js`



`/src`

- `Habit.php`
- `HabitController.php`



`/config`

- Database setup
- Helper functions



`/data`

- `habits.json` (backup)



6.1 Data persistence

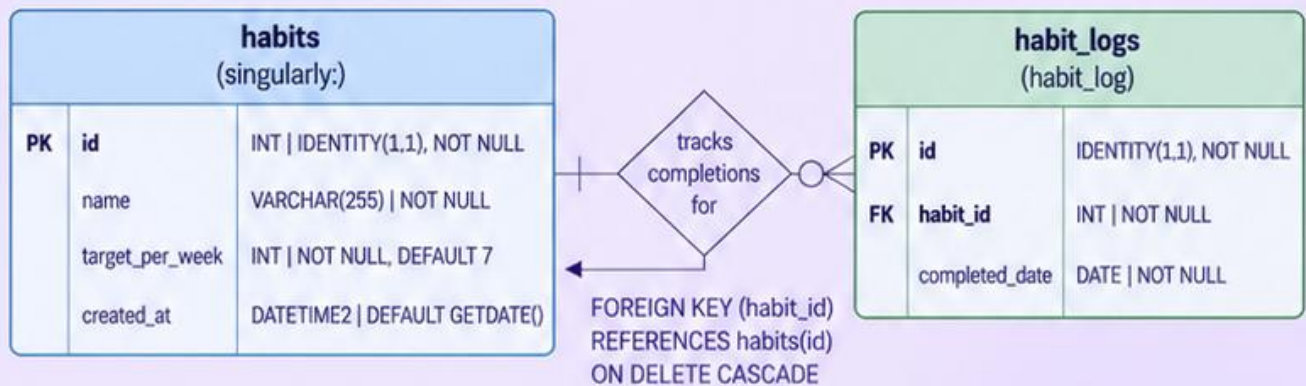
KEEPING DATA SAFE

Data persists locally inside a MySQL database managed via PHP Data Objects (PDO) for safe queries, with a secondary JSON file backup stored in `/data/habits.json`. Each user action updates the database instantly, ensuring full data persistence across browser sessions.

Los datos se persisten localmente en una base de datos MySQL administrada mediante Objetos de Datos de PHP (PDO) para consultas seguras, junto con un respaldo secundario en un archivo JSON guardado en `/data/habits.json`. Cada acción del usuario actualiza la base de datos al instante, garantizando la persistencia total entre sesiones del navegador.

Data Model Diagram

HABIT LOOP



Based on `schema.sql` definition



ORGANIZED DATA
BUILDS **REAL PROGRESS**



7. Code highlights

KEY LOGIC BEHIND THE APPLICATION

FILE AND METHOD	WHAT IT DOES	WHY IT IS INTERESTING
 HabitController:: calculateStreak()	Iterates through historical logs to count consecutive active days.	Uses PHP DateTime objects to accurately calculate streaks across different months and leap years without date bugs.
 Database:: getConnection()	Initializes a Singleton connection to MySQL via PDO.	Prevents multiple open connections by reusing a single database instance, optimizing performance and server memory.
 Habit:: saveToJson()	Encodes current habit data arrays into structured JSON text.	Implements a simple file-lock mechanism to ensure data integrity during simultaneous write operations.



Code in action

SIMPLE LOGIC
REAL IMPACT

These key methods make the habit tracker reliable, efficient, and easy to maintain, showing how simple code can solve real problems.

Code



PHP

```
1  <?php
2  class Database {
3      private static ?PDO $instance = null;
4
5      public static function getConnection(): PDO {
6          if (self::$instance === null) {
7              $serverName = "localhost\\SQLEXPRESS";
8              $database = "HabitLoopDB";
9              $username = "";
10             $password = "";
11
12             try {
13                 self::$instance = new PDO(
14                     "sqlsrv:Server=$serverName;Database=$database",
15                     $username,
16                     $password,
17                     [
18                         PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
19                         PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
20                     ]
21                 );
22             } catch (PDOException $e) {
23                 die("Connection failed: " . $e->getMessage());
24             }
25         }
26         return self::$instance;
27     }
28 }
29
```

```

1  /**
2   * HABIT LOOP - Frontend Logic
3   */
4
5  document.addEventListener('DOMContentLoaded', () => {
6      initTheme();
7      initStaggeredAnimations();
8  });
9
10 /**
11  * 1. THEME MANAGEMENT
12  */
13 function initTheme() {
14     const themeToggleBtn = document.getElementById('themeToggleBtn');
15
16     const savedTheme = localStorage.getItem('theme') || 'dark';
17     if (savedTheme === 'light') {
18         document.body.setAttribute('data-theme', 'light');
19         updateThemeButton('light');
20     } else {
21         updateThemeButton('dark');
22     }
23
24     themeToggleBtn.addEventListener('click', toggleTheme);
25 }
26
27 function toggleTheme() {
28     const currentTheme = document.body.getAttribute('data-theme');
29     let newTheme = 'dark';
30
31     if (currentTheme === 'light') {
32         document.body.removeAttribute('data-theme');
33         newTheme = 'dark';
34     } else {
35         document.body.setAttribute('data-theme', 'light');
36         newTheme = 'light';
37     }
38
39     localStorage.setItem('theme', newTheme);
40     updateThemeButton(newTheme);
41 }
42
43 function updateThemeButton(theme) {
44     const btn = document.getElementById('themeToggleBtn');
45     btn.innerHTML = theme === 'light' ? '☀️ Light' : '🌙 Dark';
46 }
47
48 /**
49  * 2. STAGGERED LIST ANIMATIONS
50  */
51 function initStaggeredAnimations() {
52     const cards = document.querySelectorAll('.habit-card');
53     cards.forEach((card) => {
54         const index = card.getAttribute('data-index');
55         if (index !== null) {
56             card.style.animationDelay = `${(parseInt(index) + 1) * 0.1}s`;
57         }
58     });
59 }
60
61 // Filtro de hábitos en tiempo real
62 document.addEventListener('DOMContentLoaded', () => {
63     const searchInput = document.getElementById('searchInput');
64
65     if (searchInput) {
66         searchInput.addEventListener('input', (e) => {
67             const searchTerm = e.target.value.toLowerCase();
68             const habitCards = document.querySelectorAll('.habit-card');
69
70             habitCards.forEach((card) => {
71                 const habitName = card.querySelector('h3').textContent.toLowerCase();
72                 if (habitName.includes(searchTerm)) {
73                     card.style.display = 'flex';
74                 } else {
75                     card.style.display = 'none';
76                 }
77             });
78         });
79     }
80 });

```

```
1 <?php
2 require_once __DIR__ . '/database.php';
3
4 class Habit {
5     private PDO $db;
6
7     // Connect to SQL Server using the Singleton database connection
8     public function __construct() {
9         $this->db = Database::getConnection();
10    }
11
12    // Fetch all habits from SQL Server ordered by creation date
13    public function getAll(): array {
14        $stmt = $this->db->query("SELECT * FROM habits ORDER BY created_at DESC");
15        return $stmt->fetchAll();
16    }
17
18    // Insert a new habit entry into the database
19    public function create(string $name, int $targetPerWeek): bool {
20        $stmt = $this->db->prepare("INSERT INTO habits (name, target_per_week) VALUES (?, ?)");
21        return $stmt->execute([$name, $targetPerWeek]);
22    }
23
24    // Remove a habit (cascades logs automatically in SQL Server)
25    public function delete(int $id): bool {
26        $stmt = $this->db->prepare("DELETE FROM habits WHERE id = ?");
27        return $stmt->execute([$id]);
28    }
29
30    // Toggle daily completion log; insert if not completed, delete if already logged
31    public function toggleLog(int $habitId, string $date): bool {
32        $stmt = $this->db->prepare("SELECT id FROM habit_logs WHERE habit_id = ? AND completed_date = ?");
33        $stmt->execute([$habitId, $date]);
34        $exists = $stmt->fetch();
35
36        if ($exists) {
37            // Remove check-in if user clicks again on the same day
38            $delete = $this->db->prepare("DELETE FROM habit_logs WHERE id = ?");
39            return $delete->execute([$exists['id']]);
40        } else {
41            // Log new completion entry
42            $insert = $this->db->prepare("INSERT INTO habit_logs (habit_id, completed_date) VALUES (?, ?)");
43            return $insert->execute([$habitId, $date]);
44        }
45    }
46
47    // Retrieve all completion dates recorded for a given habit ID
48    public function getLogs(int $habitId): array {
49        $stmt = $this->db->prepare("SELECT completed_date FROM habit_logs WHERE habit_id = ? ORDER BY completed_date DESC");
50        $stmt->execute([$habitId]);
51        return $stmt->fetchAll(PDO::FETCH_COLUMN);
52    }
53 }
54 }
```

8. Test log

TESTING FOR A BETTER EXPERIENCE

TEST
IMPROVE
SUCCEED

ID	WHAT WAS TESTED	INPUT	EXPECTED RESULT	ACTUAL RESULT	STATUS
T-01	Create new habit with valid data	Name: "Drink Water", Target: 5	Habit is saved and displayed on dashboard	Habit appeared on list instantly	PASS
T-02	Submit habit form with empty name	Name: "", Target: 3	Validation error message displayed	Error "please enter a habit name" shown	PASS
T-03	Daily check-in completion	Click check-box for "Drink Water"	Habit marked completed streak increments by 1	Streak incremented from 0 to 1	PASS
T-04	Streak calculation across months	Check-in on Aug 31 and Sep 01	Racha updates to 2 consecutive days	Racha accurately calculated as 2 days	PASS
T-05	Delete an existing habit	Click delete on "Drink Water"	Habit is removed from UI and storage	Habit successfully removed	PASS
T-06	Dark/Light theme toggle switch	Toggle ThemeToggleSwitch	UI background and font colors shift	Theme switched dynamically	PASS
T-07	Data persistence test	Refresh browser page	Previously logged habits remain intact	All data reloaded correctly from storage	PASS
T-08	Weekly progress percentage calculation	3 out of 7 days checked	Progress bar shows ~43% completion	Bar filled to 43% with exact label	PASS



Testing today

A MORE RELIABLE TOMORROW

*"Quality is not an act,
it is a habit."*

9. Challenges and conclusions

LEARN · SOLVE · IMPROVE



Challenges

PROBLEM	WHAT I SEARCHED IN ENGLISH	HOW I SOLVED IT
Streak calculation bug across months	"PHP calculate consecutive days between dates across different months"	Used native PHP DateTime diff methods instead of simple string date comparisons.
JSON file write permission error	"XAMPP PHP JSON file write permission denied local server"	Updated local directory permissions and used file_put_contents with LOCK_EX.



9.1 Conclusions

Developing Habit Loop highlighted the value of clean, modular code and accurate date handling. We combined a PHP backend with a lightweight frontend and plan to implement a database migration script and unit tests early to catch edge cases faster.

"Each challenge brings better solutions."

